

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407822
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Reisner; Samuel Joseph et al.

Method and system for optimizing bitrate selection

Abstract

A method including encoding a video program into a plurality of video streams, each of the plurality of video streams being encoded at a corresponding one of a plurality of bitrates; providing, to a plurality of viewing clients, an option to select one of the plurality of video streams; determining a streaming capacity of each of the viewing clients; and determining an improved plurality of bitrates based on streaming capacities of the plurality of viewing clients.

Inventors: Reisner; Samuel Joseph (Burbank, CA), Kidd; Michael (Burbank, CA)

Applicant: Disney Enterprises, Inc. (Burbank, CA)

Family ID: 55633737

Assignee: DISNEY ENTERPRISES, INC. (Burbank, CA)

Appl. No.: 17/248136

Filed: January 11, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210195181 A1	Jun. 24, 2021

Related U.S. Application Data

continuation parent-doc US 14508214 20141007 US 10893266 child-doc US 17248136

Publication Classification

Int. Cl.: H04L29/06 (20060101); H04N19/115 (20140101); H04N19/156 (20140101);
H04N19/162 (20140101); H04N19/184 (20140101)

U.S. Cl.:

CPC **H04N19/115** (20141101); **H04N19/156** (20141101); **H04N19/162** (20141101);
H04N19/184 (20141101);

Field of Classification Search

CPC: H04N (19/115); H04N (19/162); H04N (19/184); H04N (19/156)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6385587	12/2001	Heo	375/E7.184	G10L 19/002
6477501	12/2001	Heo	375/E7.184	G10L 19/002
6574193	12/2002	Kinrot	370/395.64	H04Q 11/0478
6618591	12/2002	Kalliokulju	455/452.2	H04W 28/22
7046805	12/2005	Fitzhardinge	348/E5.005	H04N 21/4405
7106833	12/2005	Kerpez	379/27.01	H04B 3/487
7257162	12/2006	Viscito	375/E7.199	H04N 21/242
7369613	12/2007	Concion	375/240.2	H04N 19/59
7428244	12/2007	Nilsson	375/E7.139	H04N 19/40
7881387	12/2010	Han	375/240.25	H04N 19/63
8077904	12/2010	Dixit	714/704	H04N 1/32288
8099473	12/2011	Biderman	709/224	H04N 7/17318
8259735	12/2011	Gutman	370/395.43	H04N 19/146
8265140	12/2011	Mehrotra	375/240.22	H04N 19/34
8289979	12/2011	Hiie	370/389	H04L 45/00
8325800	12/2011	Holcomb	375/240.03	H04N 19/147
8346544	12/2012	Manjunath	704/219	G10L 19/24
8406245	12/2012	Raveendran	714/776	H04L 47/38
8463929	12/2012	Hiie	709/241	H04L 45/64
8818171	12/2013	Soroushian	386/248	H04N 19/136
8838442	12/2013	Terriberry	704/205	G10L 19/0212
8942225	12/2014	Hiie	370/351	H04L 45/122
9143812	12/2014	Braness	N/A	H04N 21/6373
9154396	12/2014	Gondi	N/A	H04L 65/70
9264475	12/2015	Shivadas	N/A	H04L 65/80

9282330	12/2015	Chechik	N/A	H04N 19/115
9367288	12/2015	Wilber	N/A	G06N 3/065 H04N
9510031	12/2015	Soroushian	N/A	19/164
9654528	12/2016	Cho	N/A	H04L 65/752 H04N
9955195	12/2017	Soroushian	N/A	19/157
2002/0167607	12/2001	Eerenberg	386/E5.052	H04N 5/783 H04N
2005/0047503	12/2004	Han	375/240.12	19/619
2008/0037880	12/2007	Lai	375/240	H04N 19/64
2008/0091838	12/2007	Miceli	709/231	H04L 47/801
2010/0121974	12/2009	Einarsson	370/328	H04L 47/25 H04N
2011/0122939	12/2010	Ganesan	375/E7.026	19/115
2011/0252082	12/2010	Cobb	709/203	H04L 67/303 H04N
2012/0177101	12/2011	van der Schaar	375/E7.026	19/156
2012/0257671	12/2011	Brockmann	375/E7.126	H04N 21/64753
2013/0051767	12/2012	Soroushian	375/E7.2	H04N 21/23439
2013/0051768	12/2012	Soroushian	386/E9.011	H04N 19/154
2013/0097220	12/2012	Lyons	709/203	G07F 17/3223
2013/0170561	12/2012	Hannuksela	375/240.25	H04N 19/188
2014/0105277	12/2013	Bivolarsky	375/240.03	H04N 19/126
2014/0215536	12/2013	Maxwell	725/87	H04N 21/845
2014/0294099	12/2013	Brydon	375/240.26	H04N 19/149
2014/0325550	12/2013	Winograd	725/19	H04N 21/8456

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2012170904	12/2011	WO	H04L 65/605

OTHER PUBLICATIONS

ITU-T, H.264 Standard (Year: 2013). cited by examiner

ITU-T, H.265 Standard (Year: 2013). cited by examiner

Sullivan, G.J. (1996) “Efficient scalar quantization of exponential and Laplacian random variables”, IEEE Transactions on Information Theory. 42 (5). Institute of Electrical and Electronics Engineers (IEEE): 1365-1374 (Year: 1996). cited by examiner

Background/Summary

BACKGROUND

(1) Providers of streaming video may offer streams of the same program encoded at different bitrates, for selection either by a user or automatically by software, in order to provide a high-quality viewing experience to viewers having a variety of streaming capacities. Typically, the bitrates used for the encoding are determined and adjusted only rarely, and are based on industry-standard streaming capacities. As a result, the bitrates commonly used for encoding do not necessarily reflect increases in streaming capacity due to developing technology, or the specific capacities of the viewers of a given provider's programming or a given program.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 schematically illustrates a feedback loop for optimizing bitrate selection according to an exemplary embodiment.
- (2) FIG. 2 illustrates an exemplary chart of bandwidth usage.
- (3) FIG. 3A illustrates a first set of exemplary client performance data.
- (4) FIG. 3B illustrates a second set of exemplary client performance data.
- (5) FIG. 4 illustrates a method for optimizing bitrate selection using the feedback loop of FIG. 1 according to an exemplary embodiment.

DETAILED DESCRIPTION

- (6) Described herein are methods for optimizing the selection of bitrates used to encode streaming content. A method comprises encoding a video program into a plurality of video streams. Each of the plurality of video streams is encoded at a corresponding one of a plurality of bitrates. The method also comprises providing, to a plurality of viewing clients, an option to select one of the plurality of video streams. The method also comprises determining a streaming capacity of each of the viewing clients. The method also comprises determining an optimized plurality of bitrates based on streaming capacities of the plurality of viewing clients.
- (7) Also described herein are systems for optimizing the selection of bitrates used to encode streaming content. A system comprises a video encoder encoding a source video into a plurality of video streams. Each of the plurality of video streams is encoded at a corresponding one of a plurality of bitrates. The system also comprises an optimization logic determining a streaming capacity of each of a plurality of viewing clients of the plurality of video streams and determining an optimized plurality of bitrates based on the streaming capacities of the plurality of viewing clients. The video encoder and the optimization logic are configured to be processed by one or more processors. The one or more processors are coupled to a memory.
- (8) Also described herein is a non-transitory computer-readable storage medium storing a set of instructions that are executable by a processor. The set of instructions, when executed by the processor, causes the processor to perform operations comprising encoding a video program into a plurality of video streams. Each of the plurality of video streams is encoded at a corresponding one of a plurality of bitrates. The operations further comprise providing, to a plurality of viewing clients, an option to select one of the plurality of video streams. The operations further comprise

determining a streaming capacity of each of the viewing clients. The operations further comprise determining an optimized plurality of bitrates based on streaming capacities of the plurality of viewing clients.

(9) The exemplary embodiments may be further understood with reference to the following description and the related appended drawings, wherein like elements are provided with the same reference numerals. Specifically, the exemplary embodiments relate to methods and systems for optimizing the selection of bitrates used in encoding streaming content.

(10) It is increasingly common for consumers to view streaming video content over the Internet. Viewers of such content may have differing downstream bandwidth capacity (referred to herein as “streaming capacity” for brevity). This capacity may vary due to factors such as the type of connection (e.g., cable modem, digital subscriber line, satellite, cellular data, etc.), and, further, due to varying levels of signal quality within any given type of connection and the amount of other unrelated traffic on the connection. Throughout this disclosure, the term “streaming capacity” will be understood to refer not just to the capacity of user equipment itself, but the user's network connection and any additional interconnections between the user equipment and the source of content that is being provided to the user equipment. In some cases, the streaming capacity may not be a precise bitrate, but, rather, may refer to a range of capacities that are reasonable for providing content from the source to the user equipment. If a user attempts to view a video stream that is encoded at a bitrate that is greater than their capacity to download, the user may encounter significant buffering delays. Conversely, if a user attempts to view a video stream that is encoded at a bitrate that is less than their downloading capacity, the quality of the resulting video may be less than the user's connection can support. In either case, the user's viewing experience is worse than it would be if the user were provided with a video stream having a bitrate that closely matched the download capacity.

(11) Therefore, in order to better serve their viewers, creators and distributors of such streaming video content may offer a plurality of streams (e.g., two, five or seven streams) of a given program, with each stream encoded at a different bitrate to suit the needs of a different group of users. Users may be given the option to select a stream having a bitrate that suits their needs. As an alternative, a software application executed by the user's equipment may select an appropriate stream, or the provider or distributor may detect the quality of a user's connection and select an appropriate stream for that user.

(12) Users' capacities to stream content at greater data rates increase over time due to the evolution of technology, such as new types of connections to stream to users' homes or the introduction of higher-quality mobile data connections. Users' ability to take advantage of increasing bandwidth to realize better streaming quality depends on whether providers offer streams that are encoded at higher bitrates (and, consequently, provide higher-quality video). However, encoding settings are revisited very infrequently, often only done with the introduction of a new product or the creation of a new business arrangement with a partner, and may typically remain static for years for any given product. Further, encoding settings are typically determined based on industry-standard service levels for available products (e.g., the speed of a typical cable modem or of a typical cellular data connection), and do not take into account the capacity of a provider's actual customer base, or the capacity of a specific program's viewership.

(13) The exemplary embodiments may enable content creators and distributors to better serve their customers by adapting their encoding settings to their customers' streaming capacity. FIG. 1 conceptually illustrates an exemplary feedback loop **100** that may accomplish such adaptation. Source video **110** is received from a source by a video encoder **120**. It will be apparent to those of skill in the art that the video encoder **120** may typically be a combination of encoding software and hardware executing the encoding software according to a set of configurable encoding parameters, and that the single video encoder **120** shown in FIG. 1 may represent multiple encoders, each encompassing its own hardware and software. It will be further apparent that the source video **110**

may represent a single video program or may conceptually represent a plurality of video programs that are all subject to the processing that will be described hereinafter.

(14) The feedback loop **100** will be described with reference to a single video program but it will be apparent to those of skill in the art that the same steps may be performed in a substantially equivalent manner on a plurality of video programs. Additionally, though the exemplary embodiments will be described herein with reference to streaming video, the broader principles embodied therein are equally applicable to any type of encoded signal (e.g., audio, multimedia, etc.). Typically, the feedback loop **100**, with the exception of the viewing client **140**, may be administered by a video provider of the source video **100**. However, in other exemplary embodiments, one or more third parties may operate parts of the feedback loop **100**. In one such embodiment, the source video **100** may originate from a content generator that is different from a broadcaster encoding and providing the video streams **130**, **132** and **134** to the viewing client **140**.

(15) The video encoder **120** encodes the source video **110** into encoded video streams **130**, **132** and **134** at different encoding bitrates according to the configuration of the video encoder **120**. It will be apparent to those of skill in the art that the precise number of bitrates that are offered may vary among different embodiments and the display of three streams **130**, **132** and **134** is only exemplary. It will be further apparent to those of skill in the art that the bitrates used may vary among differing embodiments and that the specific bitrates are only to illustrate the general concepts described herein. Once the video streams **130**, **132** and **134** have been encoded, they are provided to a viewing client **140** through any of the various mechanisms known in the art, such as through a browser-based client or a dedicated software application. Similar to the source video **110**, it will be apparent to those of skill in the art that the viewing client **140** may represent a plurality of actual viewers accessing one or more streams through a plurality of different connection channels, and that only one viewing client **140** is shown for clarity.

(16) One of the streams **130**, **132** or **134** is selected for viewing by the viewing client **140**. The selection may be made by user input, or automatically by software at the viewing client **140** that selects an appropriate stream based on the connection quality available to the viewing client **140**. Based on the viewing of one of the video streams **130**, **132** or **134** by the viewing client **140**, client performance data **150** is generated. In some embodiments, client performance data **150** may be generated only upon initial selection of one of the streams **130**, **132** or **134** by the viewing client **140**; in other embodiments, further client performance data **150** may be generated if the viewing client **140** switches from one of the streams **130**, **132** or **134** to another, such as due to user request or automatically due to changing capacity of the viewing client **140**. The client performance data may include any type of data that may describe the performance of the viewing client **140**, such as streaming capacity, selection of one of the video streams **130**, **132** or **134**, connection type, etc. The client performance data **150** may be transmitted to optimization logic **160**. As described above with reference to the streams **130**, **132** and **134**, this may be accomplished through any of the various mechanisms known in the art, such as through a browser-based client or a dedicated software application. The optimization logic **160** may take the form of a codec, but in alternative embodiments may be implemented as hardware, software, firmware, or a combination of the above (e.g., a field-programmable gate array, or “FPGA”).

(17) The optimization logic **160**, like the video encoder **120**, may typically be a combination of analysis software and hardware executing the analysis software. The optimization logic **160** may analyze the client performance data **150** received across a large number of viewing clients **140** in order to optimize the encoding of the source video **110** by the video encoder **120**, and send a control signal **170** to the viewing encoder **120** to perform the optimized encoding. This analysis may involve dividing various groups of users (e.g., viewing clients **140**) into groupings (occasionally referred to by those of skill in the art as “buckets”) and selecting an encoding bitrate to target each grouping. Providing a viewer with a stream having a bitrate that is faster than the user's capacity may result in significant buffering and skipping in the user's video; conversely, a

stream having a bitrate that is less than the user's capacity may result in the user's viewing experience being worse than the user's streaming capacity could support. Thus, it may be desirable to have a large number of streams in order to provide each user (e.g., each viewing client **140**) with a stream encoded at a bitrate that is as close as possible to the bitrate that can be supported. However, because encoding a stream may entail the dedication of computational resources, and, therefore, the total resources may be directly proportional to the quantity of streams to be encoded, the number of groupings used may be equal to the number of streams that can be encoded using the amount of resources that will be devoted to the encoding process.

(18) FIG. 2 illustrates an exemplary chart **200** showing raw data plotting a range of bitrates **210**, in kilobytes per second (“kbps”), along the X axis, and a metric of use **220** along the Y axis, over a given time interval. It will be apparent to those of skill in the art that the time interval selected for analysis may be predetermined or user-configured, and may vary among different implementations of a system implementing the feedback loop **100**. Consideration of too recent a time window prior to the time of analysis may result in an analysis that considers short-term fluctuations in user bandwidth capacity. Conversely, consideration of too long a time window prior to the time of the analysis may result in an analysis that is not responsive to changes in transmission technology. The time range may be anywhere from a few hours in length (e.g., the time between two consecutive event starts) to a month, and may typically be in the range of one day to one week.

(19) In the exemplary chart **200**, the metric shown is a number of bandwidth tests (e.g., pings) over the time interval. However, it will be apparent to those of skill in the art that different metrics may be used without departing from the broader principles from the exemplary embodiments. For example, alternative metrics may include a number of users or a number of hours of viewed content. The optimization logic **160** may consider data such as shown in the exemplary chart **200**; divide the metrics, which are representative of the proportion of users having a given bitrate capacity, into groupings as described above, and select an encoding bitrate for each grouping.

(20) Considering the specific data shown in the chart **200**, the optimization logic may identify a first grouping containing users at bitrates up to point **230**, a second grouping containing users at bitrates between points **230** and **232**, a third grouping containing users at bitrates between points **232** and **234**, a fourth grouping containing users at bitrates between points **234** and **236**, and a fifth grouping containing users at bitrates greater than point **236**. The optimization logic **160** may then determine an optimal set of encoding bitrates such that one of the determined bitrates is optimal for each of the groupings.

(21) In one exemplary embodiment, the bitrate selected for each of the groupings is the bitrate of the user with the slowest streaming capacity in each of the groupings; in another exemplary embodiment, the bitrate selected for each grouping is the average of the streaming capacity of all the users in the grouping. There may be a variety of other statistical or empirical ways for selecting a most appropriate bitrate for the plurality of users contained within a grouping. In one embodiment, the bitrate selected for a grouping may be a bitrate that is lower than the lowest capacity of any of the users within the grouping to ensure that all of the users in the group are capable of operating at the selected bitrate. Returning to the example of FIG. 2, a bitrate corresponding to point **240** may be chosen for the first grouping, a bitrate corresponding to point **242** may be chosen for the second grouping, a bitrate corresponding to point **244** may be chosen for the third grouping, a bitrate corresponding to point **246** may be chosen for the fourth grouping, and a bitrate corresponding to point **248** may be chosen for the fifth grouping. These selections are only exemplary and other divisions of users into groupings and selection of bitrates for groupings may be made for the same or different data.

(22) In one exemplary embodiment, the client performance data **150** may take the form of a number of users selecting each of a plurality of streams of the same program encoded at different bitrates, or, alternatively, a number of hours of video viewed over each of a plurality of streams. FIGS. 3A and 3B illustrate different result sets of client performance data **150**. FIG. 3A shows a bar

graph **300** of data for five different streams: a first stream **310** encoded at a bitrate of 480 kbps, a second stream **312** encoded at a bitrate of 1000 kbps, a third stream **314** encoded at a bitrate of 1500 kbps, a fourth stream **316** encoded at a bitrate of 2120 kbps, and a fifth stream **318** encoded at a bitrate of 2940 kbps. The number of hours viewed for each of the streams **310**, **312**, **314**, **316** and **318** over a selected time interval is shown along the Y axis **320**.

(23) FIG. 3A shows result data **330** for stream **310** indicating roughly 13,000 hours viewed during the time interval. Result data **332** for stream **312** shows roughly 4,000 hours viewed during the time interval. Result data **334** for stream **314** indicates roughly 2,500 hours viewed during the time interval. Result data **336** for stream **316** shows roughly 2,000 hours viewed during the time interval, and result data **338** for stream **318** indicates roughly 4,500 hours viewed during the time interval. Because of the large number of viewer devices selecting stream **310**, the optimization logic **160** may opt to provide an additional low-end bitrate to provide a viewing experience more tailored to the needs/preferences of these viewers. For example, the optimization logic **160** may eliminate stream **318** (which might result in the viewers who selected stream **318** selecting stream **316** instead) and create a new stream at a lower bitrate than that of stream **310**. Further client performance data **150** may be generated based on this new set of streams, and the optimization logic **160** may analyze this further client performance data **150** and further adjust the bitrates, in keeping with the typical manner of operation of a feedback loop.

(24) FIG. 3B shows an alternate bar graph **350** showing different results for the same five streams **310**, **312**, **314**, **316** and **318** shown in FIG. 3A along a differently-scaled Y axis **360**. Result data **370** for stream **310** indicates roughly 180,000 hours viewed during the time interval. Result data **372** for stream **312** shows roughly 120,000 hours viewed during the time interval. Result data **374** for stream **314** indicates roughly 80,000 hours viewed during the time interval. Result data **376** for stream **316** shows roughly 120,000 hours viewed during the time interval. Result data **378** for stream **318** indicates roughly 700,000 hours viewed during the time interval. Because of the large number of viewers selecting stream **318**, including more viewers than selected the other four streams combined, the optimization logic may opt to provide an additional high-end bitrate to provide a viewing experience more tailored to the needs/preferences of these viewers. For example, the optimization logic **160** may eliminate stream **316** (which might result in the viewers who selected stream **316** selecting stream **314** instead) and create a new stream at a faster bitrate than that of stream **318**. Further client performance data **150** may be generated based on this new set of streams, and the optimization logic **160** may analyze this further client performance data **150** and further adjust the bitrates, in keeping with the typical manner of operation of a feedback loop. The specific possible actions by the optimization logic **160** described above are only exemplary, it will be apparent to those of skill in the art that there may be a variety of optimization algorithms known in the art that might be applied to the client performance data **150** by the optimization logic **160** at this point.

(25) As described above, because each additional stream offered to users has a cost in terms of the requirement to devote computing resources to encode the stream, the provider of the streams may limit the number of bitrates (and, correspondingly, streams) that are available. For example, a provider may determine that it wishes to allocate computing resources sufficient to encode three streams. However, the limited quantity of streams offered to users to choose from may result in less useful data about the various users' streaming capacity. Therefore, in one exemplary embodiment, users may be offered streams at bitrates that have not actually been encoded in order to count users that select those streams in addition to the streams that have been encoded. It will be apparent to those of skill in the art that, in the context of selection, streams may be offered to users in a variety of manners, including providing users with the opportunity to select from all available streams, providing users with the opportunity to select only from a subset of streams that are suitable for the individual users' capacities, providing data to be used by user equipment in making an automatic selection of an appropriate stream, or any other manner of providing streaming data at different

bitrates that is known in the art.

(26) For example, referring to the feedback loop **100** of FIG. **1** and the data sets shown in FIGS. **3A** and **3B**, the video encoder **120** may initially be configured to encode streams **130**, **132** and **134** at 480 kbps (e.g., stream **310**), 1500 kbps (e.g., stream **314**), and 2940 kbps (e.g., stream **318**). However, to enhance the precision (and, thus, the usefulness) of the client performance data **150**, fictitious bitrates may be created. For example, stream **312** at 1000 kbps, interspersed between stream **310** at 480 kbps and stream **314** at 1500 kbps, may be created based on stream **310** and offered to users for selection. Such streams (e.g., stream **312** at 1000 kbps, interspersed between stream **310** at 480 kbps and stream **314** at 1500 kbps) are created by padding a lower bitrate stream with null data to create a higher bitrate stream for transmission to a viewing client **140**. The padding may be performed server-side (e.g., as an additional function performed by video encoder **120**) or by a content delivery network (“CDN”) in order not to have the null data be transmitted by the broadcaster. Thus, by offering more bitrates than are actually encoded, the feedback loop **100** may generate more useful client performance data **150**.

(27) FIG. **4** illustrates an exemplary method **400** for implementing the feedback loop **100** of FIG. **1**. The method **400** will be described herein with reference to the elements of the exemplary feedback loop **100**. In step **410**, initial bitrates at which the source video **110** is to be encoded by the video encoder **120** are selected. The initial bitrates may be determined, for example, using prior art methods such as based on industry-level averages for users' downstream speed capacity, based on competitors' streaming offerings, etc. In step **420**, the source video **110** is encoded by the video encoder **120** at the bitrates selected in step **410**.

(28) In step **430**, any fictitious bitrates that may be offered to user devices for selection are selected. Fictitious bitrates may be selected either algorithmically or through manual input from a system operator. In one algorithmic embodiment, one or more fictitious bitrates can be selected at a point or points between real bitrates. This selection can be at one predetermined point in the range (e.g., at the midpoint of the range; for real bitrates 100 kb/s and 200 kb/s, this would select a fictitious bitrates of 150 kb/s), at two predetermined points (e.g., at 30% of the way through the range and 50% of the way through the range; for real bitrates 100 kb/s and 200 kb/s, this would select fictitious bitrates of 130 kb/s and 150 kb/s), or at a randomized point in the range (e.g., for real bitrates 100 kb/s and 200 kb/s, a fictitious bitrate could be randomly chosen anywhere within the range, with random selection being performed through any randomization means known in the art).

(29) In another algorithmic embodiment, a fictitious bitrate may be selected at a predetermined or randomized percentage higher than each real bitrate, provided that this increase does not exceed the next higher bitrate. This technique may be particularly suitable for consideration of rates higher than the highest real bitrate. For example, a predetermined percentage of 5% may be used (e.g., for a real bitrate of 100 kb/s, a fictitious bitrate of 105 kb/s may be created). Multiple bitrates may also be selected in this manner (e.g., fictitious bitrates created at 5% and 10% increases over each real bitrate).

(30) In another algorithmic embodiment, a fictitious bitrate may be selected at a predetermined or randomized amount higher than each real bitrate, provided that this increase does not exceed the next higher bitrate. This technique may also be particularly suitable for consideration of rates higher than the highest real bitrate. For example, a predetermined increase of 20 kb/s may be used (e.g., for a real bitrate of 100 kb/s, a fictitious bitrate of 120 kb/s may be created). Multiple bitrates may also be selected in this manner (e.g., fictitious bitrates created at 10 kb/s and 20 kb/s increases over each real bitrate).

(31) In step **440**, streams at desired fictitious bitrates that were identified in step **430** are created based on streams that were encoded in step **420**. As described above, fictitious bitrates may be employed in order to obtain user selection data about a wider variety of bitrate options than a broadcaster may wish to provide, due to the resource cost of providing multiple bitrates. Also as

described above, streams at fictitious bitrates may be created by padding a lower-bitrate stream with null (e.g., dummy) data. The selection and creation of fictitious bitrate streams are not required because the method **400** may still provide useful data relating to user preferences without employing fictitious bitrates. Additionally, null data is only one example of a type of data that may be used as padding to generate a stream at a fictitious bitrate. In another embodiment, rather than null data, other non-video data or separate video data may be used as padding. In a further embodiment, random noise data could be used in order to prevent the effects of transparent compression systems within the data stream.

(32) In step **450**, the viewing client **140** may be provided with an option to select from the various available bitrates. This may include streams at both real bitrates, as encoded in step **420**, and streams at fictitious bitrates, as generated in step **440**. The selection by the viewing client **140** may be manual (e.g., a user of the viewing client **140** may input a selection) or automatic (e.g., a software application executed at the viewing client **140** may determine an appropriate stream for the viewing client **140** based on its streaming capacity). It will be apparent to those of skill in the art that the streams are, in actuality, provided to a plurality of viewing clients in order to yield useful client performance data **150**, and that a single viewing client **140** is discussed herein for purposes of clarity. Once the viewing client **140** makes a selection, the selected stream may be provided to the viewing client **140** from the provider through known means, such as via a CDN, but the actual provision of the selected stream is beyond the scope of the exemplary embodiments and will not be discussed further herein. As previously noted, the selection by the viewing client **140** may be made manually by a user of the viewing client **140**, or automatically by software of the viewing client **140** based on the streaming capacity of the viewing client **140**. Additionally, automatic switching between streams at different bitrates may be done “on the fly” during the course of a program, not just at the beginning of a particular program.

(33) The selection of one of the streams **130**, **132** and **134** by the viewing client **140** is only one way to determine the streaming capacity of the viewing client **140**. In another exemplary embodiment, a software application executed by the viewing client **140** may determine the streaming capacity of the viewing client **140** and report the capacity to the optimization logic **160** as the client performance data **150**. There may be a variety of ways in which such a software application may make this determination, such as by downloading a file of known size and determining streaming capacity by dividing the size by the download time.

(34) In step **460**, client performance data **150** is sent from the viewing client **140** to the optimization logic **160**. As noted above, this may be accomplished by means of a browser-based client, a dedicated software application, or any other appropriate mechanism that is known in the art. Alternatively, in some embodiments, no affirmative “sending” step may be required, but, rather, the client performance data **150** may be generated provider-side based on the sending of the different streams to various users. It will be apparent to those of skill in the art that, from the perspective of the optimization logic **160**, client performance data **150** may originate from a plurality of viewing clients that are represented schematically by the viewing client **140**. In step **470**, the optimization logic **160** aggregates client performance data **150** received from a plurality of viewing clients to yield statistical data such as that shown in FIGS. 2, 3A and 3B.

(35) In step **480**, the optimization logic **160** determines changes to be made to the encoding bitrates, if any, based on the client performance data **150**. As described above, this may involve eliminating a bitrate selected by few users, creating a new bitrate to replace an eliminated bitrate, or use of other optimization methodology known in the art. In step **490**, the optimization logic **160** sends a control signal **170** to the video encoder **120** containing instructions for changes to the bitrates used in the encoding. Following step **490**, the method **400** returns to step **420**, and the video encoder **120** uses the newly-determined bitrates for subsequent encoding. It will be apparent to those of skill in the art that the method **400** may typically be an ongoing optimization process without a predefined endpoint, and therefore does not include a termination point, but that

performance of the optimization process may be terminated by the content provider at any desired point in time. It will be further apparent to those of skill in the art that an operator of the feedback loop may have the option to limit the frequency of updating the bitrates, or of the performance of the method **400** as a whole, in order to minimize computing demands.

(36) The exemplary embodiments may provide for a dynamic and updated selection of bitrates for the encoding of streaming video. As opposed to prior art methods that may simply base encoding bitrates on industry standard technology levels, the exemplary embodiments may provide an initial set of bitrates based on industry standards or a similar basis, and may then continually optimize the encoding bitrates based on the streaming capacities of a provider's viewers. Although the exemplary embodiments have been described above with reference to the determination of "optimal" bitrates, it will be apparent to those of skill in the art that a completely optimal result may be an unattainable goal, and that, even if the exemplary embodiments merely provide an improved set of bitrates that may not be optimal in an absolute sense, this may still be a valuable result. The exemplary embodiments may provide for bitrates that evolve along with developments in streaming technology on a more regular basis than prior techniques. The exemplary embodiments may further enable the provider's streaming offerings to best cater to the provider's specific viewer base, or even to the viewers of a specific program. As a result, more viewers of the provider's programming may enjoy an optimized viewing experience.

(37) Those of skill in the art will understand that the above-described exemplary embodiments may be implemented in any number of matters, including as a software module, as a combination of hardware and software, etc. For example, the exemplary method **400** may be embodied in a program stored in a non-transitory storage medium and containing lines of code that, when compiled, may be executed by a processor.

(38) It will be apparent to those skilled in the art that various modifications may be made to the exemplary embodiments, without departing from the spirit or the scope of the invention. Thus, it is intended that the present invention cover modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. A method, comprising: encoding each of a first plurality of video streams of a video program at one of a first plurality of predetermined bitrates; generating a first alternate video stream at a first alternate bitrate that is not any of the one of the first plurality of the predetermined bitrates at which the first plurality of video streams of the video program are encoded, wherein the first alternate bitrate is selected at a randomized percentage higher than one of the first plurality of the predetermined bitrates but not exceeding a bitrate of a second one of the first plurality of the predetermined bitrates that is a next higher bitrate from the one of the first plurality of the predetermined bitrates; providing, to a plurality of viewing clients, an option to select one of the first plurality of video streams or the first alternate video stream at the first alternate bitrate; and determining, by an optimization logic, a second plurality of bitrates based on at least one selection of a first alternate stream at the first alternate bitrate by the plurality of viewing clients.
2. The method of claim 1, wherein the first alternate video stream at the first alternate bitrate is generated based on padding one of the first plurality of video streams with additional data.
3. The method of claim 2, wherein the additional data is one of null data, random noise, video data or non-video data and wherein the first alternate video stream at the first alternate bitrate is higher than the bitrate of the one of the first plurality of video streams that is selected to generate the first alternate video stream at the first alternate bitrate.
4. The method of claim 1, further comprising encoding each of a second plurality of video streams using a corresponding one of the second plurality of bitrates.
5. The method of claim 1, further comprising: receiving an indication of a streaming capacity from

client-side equipment of each of the plurality of viewing clients.

6. The method of claim 5, wherein the second plurality of bitrates is further based on the streaming capacity.

7. The method of claim 1, wherein the selection of the first alternate stream is a user selection.

8. The method of claim 1, wherein the selection of the first alternate stream is a software-based selection.

9. A system, comprising: a video encoder configured to encode a first plurality of video streams of a video program, each video stream being encoded at one of a first plurality of predetermined bitrates and generate a first alternate video stream at a first alternate bitrate that is not any of the one of the first plurality of the predetermined bitrates at which the first plurality of video streams of the video program are encoded, wherein the first alternate bitrate is selected at a randomized percentage higher than one of the first plurality of the predetermined bitrates but not exceeding a bitrate of a second one of the first plurality of the predetermined bitrates that is a next higher bitrate from the one of the first plurality of the predetermined bitrates; and an optimization logic configured to determine a second plurality of bitrates based on at least one selection of the first alternate video stream at the first alternate bitrate by one of a plurality of viewing clients, wherein the video encoder and the optimization logic are configured to be processed by one or more processors.

10. The system of claim 9, wherein the first alternate video stream at the first alternate bitrate is generated based on padding one of the first plurality of video streams with additional data.

11. The system of claim 10, wherein the additional data is one of null data, random noise, video data or non-video data and wherein the first alternate video stream at the first alternate bitrate.

12. The system of claim 9, wherein the first alternate video stream at the first alternate bitrate is higher than the bitrate of the one of the first plurality of video streams that is selected to generate the first alternate video stream at the first alternate bitrate.

13. The system of claim 9, wherein the video encoder is further configured to encode a second plurality of video streams with one of the second plurality of bitrates.

14. The system of claim 9, wherein the optimization logic is further configured to receive an indication of a streaming capacity from client-side equipment of each of the plurality of viewing clients.

15. The system of claim 14, wherein the second plurality of bitrates is further based on the streaming capacity.

16. A non-transitory computer readable storage medium storing a set of instructions that are executable by a processor, the set of instructions, when executed by the processor, causing the processor to perform operations comprising: encoding a first plurality of video streams of a video program, each video stream being encoded at one of a first plurality of predetermined bitrates; generating a first alternate video stream at a first alternate bitrate that is not any of the one of the first plurality of the predetermined bitrates at which the first plurality of video streams of the video program are encoded, wherein the first alternate bitrate is selected at a randomized percentage higher than the highest bitrate of the first plurality of the predetermined bitrates; providing, to a plurality of viewing clients, an option to select one of the first plurality of video streams or the first alternate video stream at the first alternate bitrate; and determining, by an optimization logic, a second plurality of bitrates based on at least one selection of the first alternate video stream at the first alternate bitrate by the plurality of viewing clients.

17. The non-transitory computer readable storage medium of claim 16, wherein the first alternate video stream at the first alternate bitrate is generated based on padding one of the first plurality of video streams with additional data.

18. The non-transitory computer readable storage medium of claim 17, wherein the additional data is one of null data, random noise, video data or non-video data and wherein the first alternate video

stream at the first alternate bitrate is higher than the bitrate of the one of the first plurality of video streams that is selected to generate the first alternate video stream at the first alternate bitrate.
